# Ethan Wu

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## Profile

Dedicated and analytical 3rd-year Mechatronics and Robotics Engineering student at Queen's University with a strong foundation in embedded systems, high-level programming, CAD, and automated workflow design. Proven leadership experience managing the Automation Teams, developing robotics software stacks (C++, ROS2) for competitive design challenges, and coordinating multidisciplinary university projects.

## Education

**Queen's University** Kingston, ON
*Bachelor of Applied Science in Mechatronics and Robotics Engineering (GPA: 3.89)* *Sept 2024 – Expected May 2028*
- Relevant Coursework: Sensors and Electric Actuators, CAD, Electronics, Signals and Systems, Microprocessor Interfacing, Embedded Systems, Computer Architecture, Feedback Control Systems, Manufacturing Processes, Corporate Law.

**University of Toronto Schools (UTS)** Toronto, ON
*High School Diploma (Grade Average: 97%)* *Sept 2018 – June 2024*
- Awards: AP Scholar Award, Ontario Scholar, Honor Roll (2019–2024).

## Leadership & Extracurricular Experience

**Task Subteam Lead** Sept 2026 – Present
*Queen's Lunar Exploration Team (NASA HERC)* *Kingston, ON*
- Spearhead multidisciplinary integration (mechanical, electrical, and software) for the 2027 NASA HERC autonomous task mechanism, managing RF command parsing and real-time target data acquisition.
- Architect computer vision pipelines utilizing Raspberry Pi and OpenCV to isolate shape targets, apply dynamic digital color filters, and calculate rover-to-target distance via focal geometry.
- Design and validate robust hardware architectures and C++/ROS2 control software stacks to streamline rover telemetry and actuator response.

**Automation Team Manager (formerly Project Lead)** Oct 2025 – Present
*Engineering Society at Queen's University* *Kingston, ON*
- Direct recruitment, developer onboarding, and project management activities for the Engineering Society automation team (approximately 20+ members).
- Created Power Automate flows to track training completions and streamline hiring workflows; collaborated with the Director of HR and VPFA to design and optimize future hiring protocols.
- Managing projects requested by Clark Hall Pub (real-time ordering queue), Campus Equipment Outfitters (dynamic merchandise ordering website), Director of Design Teams (an engineering extracurricular catalogue), etc..

**Finance Coordinator** Jul 2026 – Present
*Queen's Space Conference* *Kingston, ON*
- Managed financial logistics, budgeting, and sponsorship coordination for the annual space conference.

**Challenges Coordinator** Jun 2026 – Present
*Queen's Engineering Competition* *Kingston, ON*
- Designed, drafted, and budgeted the Senior Design Challenge for QEC 2026, establishing technical evaluation criteria.
- Validated proper electronic and mechanical materials via testing and prototyping to teams to ensure challenge feasibility in a 6-hour hackathon-style competition.

**Embedded Systems General Member** Sept 2025 – Jun 2026
*Queen's AutoDrive Team* *Kingston, ON*
- Won 2nd place for the Mobility Innovation Challenge in the SAE AutoDrive Competition by proposing "Learning Mode" for new drivers using autonomous vehicles.
- Developed a C++ memory monitoring module for an embedded system watchdog to detect memory leaks and prevent system crashes.
- Researched methods to extract battery telemetry data from legacy hardware applications.

**Avionics General Member** Sept 2025 – Apr 2026
*Queen's Rocket Engineering Team (QRET)* *Kingston, ON*
- Contributed to avionics circuit design, sensor integration, and telemetry systems for high-power rocketry.

**QSuccess Mentor** Jun 2025 – Apr 2026
*Queen's University* *Kingston, ON (Remote)*
- Supported incoming students throughout their first academic year, assisting mentees in establishing academic goals and fostering a growth mindset through bi-weekly mentoring sessions.

**First-Year Project Coordinator** Oct 2024 – Apr 2025
*Queen's Rocket Engineering Team (QRET)* *Kingston, ON*
- Drafted newsletters for sponsors and alumni; reviewed sponsorship and bursary documents; researched financial initiatives.

**Engineering Orientation Leader (ACTION)** Mar 2025 – Sep 2025
*Queen's University Faculty of Engineering* *Kingston, ON*
- Welcomed and mentored incoming first-year engineering students, creating an energetic, supportive environment and leading large-scale campus events.

**VEX Robotics Team Senior Executive** Sept 2023 – Jun 2024
*University of Toronto Schools (UTS)* *Toronto, ON*
- Trained 50+ members on robot design, CAD (SolidWorks/Fusion 360), prototyping, and mechanical/electrical assembly.
- Helped lead the team and delegate tasks at regional and international championships, including the VEX Robotics World Championship 2024.

**Stage Crew Senior Executive & Member** Sept 2021 – Jun 2024
*University of Toronto Schools (UTS)* *Toronto, ON*
- Elected executive member managing a 50+ member crew; mentored junior members on operating lighting and sound technology.
- Operated audio, lighting, and visual tech during large-scale live performances under strict time constraints.

## Engineering Projects

**Autonomous Pollution Monitoring Device (TriPAD)** | *C++, ESP32, Sensors, Autodesk Fusion* Apr. 2026 – Aug. 2026
- Built and calibrated a remote environmental monitoring system combining an ESP32, PMS5003 particulate sensor, MAX9814 microphone, and MPU6050 IMU.
- Configured low-level I2C communication protocols, custom firmware logic, and data-logging pipelines to record site sound, dust, and vibration levels.
- Designed a custom 3D-printed Autodesk Fusion enclosure and secured 3rd place in the General's Lab Innovation Project (GLIP) competition.

**Prototype Launcher Turret** | *Arduino Uno, C++, Sensor Shields, Motor Boards, Servos* Oct 2026
- Designed and built a high-velocity pan-tilt BB launching turret utilizing DC motors, SG90/MG996R servos, DFRobot IO Expansion Shield v5, and DF-MD v1.3.
- Programmed Arduino Uno in C++ for closed-loop motor positioning and sensor-feedback control routines.
- Engineered custom structural mounting using everyday materials and 3D-printed parts in Autodesk Fusion.

**Autonomous Transforming Robot (In-Progress)** | *Arduino Uno, C++, Sensor Shields, Motor Boards, Servos* Oct 2026
- Designing a 30+ DOF kinematic assembly with interference checks, motion analysis, and tolerance tuning for 3D printing.

**AI in Criminal Justice Research Paper** | *UTS Capstone Project* 2023 – 2024
- Investigated AI integration in criminal justice, analyzing ethical implications regarding privacy, accuracy, and discrimination.
- Conducted survey-based research assessing adolescent perceptions of algorithmic justice to inform stakeholders.

## Professional & Employment Experience

**Teaching Assistant – APSC 101** Aug 2026 – Present
*Stephen J. R. Smith Faculty of Engineering and Applied Science, Queen's University* *Kingston, ON*
- Providing in-class academic support, troubleshooting, and design guidance for first-year engineering students in "APSC 101: Problem Analysis and Modeling."

**Line Cook** Jun 2025 – Aug 2025
*Pearl Diver* *Toronto, ON*
- Executed fast-paced kitchen operations, maintaining food safety standards and working efficiently under pressure in a high-volume restaurant environment.

**Document Clerk** May 2024 – Aug 2024
*Houghton, Sloniowski & Stengel (formerly Youthfully)* *Toronto, ON*
- Assisted with the preparation of numerous family law documents (including Applications, Affidavits, and Financial Statements) in compliance with legislation.
- Conducted legal case research and drafted internal legal memos to assist with active client files, sharpening extreme attention to detail.

**Kitchen Assistant / Summer Camp Counselor** Jun 2019 – Aug 2023
*The Chef Upstairs* *Vaughan, ON*

- Supervised and delegated tasks to campers aged 6–17 in cooking, cleanup, and workstation coordination, consistently delivering meals on schedule.
- Handled menu-planning, kitchen/pantry organization, inventory control, sanitation, and provided first aid when needed.

**Interviewer and Surveyor** 2019 – 2021
*Angel Alert* *Toronto, ON*
- Conducted structured interviews with elderly community members regarding personal safety, technology adoption, and social media habits.

**Leaflet Distributor** 2018 – 2020
*Ontario Clean Air Alliance* *Toronto, ON*
- Strategically planned optimal door-to-door distribution routes, completing 3+ hours of active fieldwork daily and delivering literature to over 1,200 households.

## Athletics & Community Involvement

**Soccer Player (Goalie & Defense)** 2011 – 2018
*Milliken Soccer Club* *Markham, ON*
- Progressed from recreational level through house league and competitive rep teams over an 8-year span.

## Certifications

- **Standard First Aid & CPR** – University of Toronto Schools (2024 – 2027)
- **Smart Serve** – Smart Serve Ontario (2025 – 2030)
- **WHMIS Certification**

## Technical Skills

**Software & Controls**: C/C++, Python, Java, Reinforcement Learning (RL), ROS2, Nav2, SLAM, OpenCV, PID/PI Control, Sensor Fusion, Kinematics, SQL, Linux, HTML, CSS
**Hardware & Embedded**: ESP32, Raspberry Pi, Arduino, VHDL, CAN bus, UART, I2C, PMS5003, MAX9814, MPU6050, LiDAR, Encoders
**CAD & Tools**: Autodesk Fusion, SOLIDWORKS, VHDL, LTSpice, Quartus, MATLAB, MS Power Automate, Git, LaTeX
**Hands-On**: Soldering, Circuit Debugging, Machining, Welding, Woodworking, 3D Printing
**Languages**: Cantonese, Mandarin, French